



Technical reference manual
1.0.0

EtherBench

Peripherals

Offset	Size (Bytes)	Description
0x00	2	Peripheral config marker, constant to 0xEBB2
0x02	2	Page CRC
0x04	2	Enabled peripherals (see mask at the bottom)
0x06	1	GPIO Direction (4 LSB) : 1 : in; 0 : out
0x07	1	GPIO Value (4 LSB)
0x08	3	UART 0 Config
0x0B	3	UART 1 Config
0x0E	3	UART 2 Config
0x11	3	SPI Config
0x15	4	CAN Config
0x19	4	JTAG / SWD Config
0x1D	3	I2C Config

Peripherals flags

Bit	Name
0	UART 0
1	UART 1
2	UART 2
3	Reserved (constant to 0)
4	I3C / I2C

5	SPI
6	CAN
7	JTAG
8	Reserved (constant to 0)
9	GPIO 0
10	GPIO 1
11	GPIO 2
12	GPIO 3
13	Reserved (constant to 0)
14	Reserved (constant to 0)
15	Reserved (constant to 0)

UART Config

Offset	Size	Description
0x00	2	Speed : Baudrate / 100
0x02	1	Bits 0-1 : Transfer size : 7 + 2'bXX. Bits 3-4 : Stop bits number : 2'bXX (1 or 2). Bit 7 : Enable parity.

I2C Config

Offset	Size	Description
0x00	1	Speed : Baudrate / 1000
0x01	1	Bit 0 : Enable slave mode. Bit 2 : Enable 10 bit address. Bits 7-5 : Slave address [10-8]

0x02	1	Slave address [7:0]
------	---	---------------------

SPI Config

Offset	Size	Description
0x00	2	Speed : Baudrate / 1000
0x02	1	Transfer size (4-32)

CAN Config

Offset	Size	Description
0x00	1	???
0x01	1	???
0x00	2	???
0x01	3	???

JTAG / SWD Config

Offset	Size	Description
0x00	2	Speed : Baudrate / 1000
0x02	2	Bit 0 : Connect under reset

Yaml arguments

When using the automated page builder, available on the EtherBenchProbe repo, each bus is configured by it's own key. Therefore, are searched:

- uart0
- uart1

- uart2
- spi
- can
- jtag or swd
- i2C

The present of any key will trigger the build of the page. Missing keys are going to be filled with the default value. For each of them, the following key are searched.

UART

Key	Format	Default value
baudrate	Integer	115200
size	Integer	8
stop bits	Integer	1
partity	Integer	0

I2C

Key	Format	Default value
baudrate	Integer	115200
slave address	Integer	0
slave mod	Integer	0

SPI

Key	Format	Default value
baudrate	Any SI value (1MHz, 16MHz)	"1MHz"
size	Integer	8

CAN

Key	Format	Default value
baudrate	Any SI value (1MHz, 16MHz)	"1MHz"
size	Integer	8

JTAG/SWD

Key	Format	Default value
baudrate	Any SI value (1MHz, 16MHz)	"1MHz"
size	Integer	8